Experimental Simulations

Pre-Test Report

by

Group 29

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0. Introduction

The primary challenge of this experiment is that the scale model's propellers are not equipped with rotating shaft balances, preventing direct thrust measurement. Instead, the external balance records only the net longitudinal force, which is a superposition of airframe drag and propeller thrust. Consequently, the main objective is to isolate the airframe's aerodynamic coefficients by estimating thrust using Blade Element Momentum (BEM) simulations. This approach allows for the characterization of longitudinal stability, elevator effectiveness, and acoustic performance, specifically during the negative thrust (braking) regime used for approach.

1. Description of Performance Indicators and Dimensional Analysis

To ensure that experimental data from the wind-tunnel model can be scaled to represent the full-scale aircraft, a systematic dimensional analysis is performed. This process reduces the large number of physical variables into a smaller set of dimensionless parameters using the Buckingham Π-theorem. This chapter outlines this derivation and defines the key performance indicators for the experiment.

1.1. Dimensional Analysis of Forward Flight

1.1.1. Relevant Variables and their Dimension Functions

The first step is to identify all physical variables influencing the aircraft's performance in forward flight and their corresponding dimensions in the Mass (M), Length (L), and Time (T) system:

- Thrust (T): $[MLT^{-2}]$
- Freestream density (ρ_∞): $[ML^{-3}]$
- Freestream velocity (V_∞): $[LT^{-1}]$
- Speed of sound (a_∞): $[LT^{-1}]$
- Freestream viscosity (μ_∞): $[ML^{-1}T^{-1}]$
- Propeller diameter (D): $[L]$
- Rotational frequency (n): $[T^{-1}]$
- Number of blades (N_B): [1]
- Blade pitch angle (β): [1]

1.1.2. Independent Scaling Variables

According to the Buckingham Π-theorem, we choose a number of repeating variables equal to the number of base dimensions ($k = 3$). To analyze the aircraft in forward flight, the most logical set is: Density (ρ_∞), Freestream Velocity (V_∞) and Propeller Diameter (D)

1.1.3. Dimensionless Parameters (Π-Products)

The remaining (dependent) variables are normalized by the chosen independent variables to create a set of dimensionless groups (Π-products). The physical relationship is thus simplified from $T = f(\rho_\infty, V_\infty, ...)$ to a function of these dimensionless products. Each Π-product and its physical meaning is defined below:

- $\Pi_1 = \frac{T}{\rho_\infty V_\infty^2 D^2}$: This is the **Thrust Coefficient** (T_C), relating the propeller thrust to the freestream dynamic pressure.
- $\Pi_2 = \frac{a_\infty}{V_\infty}$: This is the inverse of the **Freestream Mach Number** ($Ma = V_\infty / a_\infty$), which governs compressibility effects.
- $\Pi_3 = \frac{\mu_\infty}{\rho_\infty V_\infty D}$: This is the inverse of the **Freestream Reynolds Number** ($Re = \rho_\infty V_\infty D / \mu_\infty$), which governs viscous effects.
- $\Pi_4 = \frac{nD}{V_\infty}$: This is the inverse of the **Advance Ratio** ($J = V_\infty / nD$), the key parameter for ensuring propeller kinematic similarity.
- $\Pi_5 = N_B$ and $\Pi_6 = \beta$: These represent the dimensionless geometric parameters of the propeller.

1.2. Performance Indicators and Similarity Parameters

The dimensional analysis provides a formal basis for defining the dimensionless coefficients used to evaluate the aircraft's performance and the key parameters that ensure the experiment is scalable.

1.2.1. Governing Similarity Parameters

To achieve dynamic similarity, the following dimensionless parameters must be considered.

- **Reynolds Number (Re):** This parameter will not be matched to full-scale flight conditions. Since the experiment uses a sub-scale model in an atmospheric wind tunnel, fluid density (ρ) and viscosity (μ) are fixed. To compensate for the smaller model length (L) and match the full-scale Reynolds number, the freestream velocity (V) would need to be increased to an impractically high, supersonic level. This is not feasible due to the facility's maximum speed and the structural limits of the model, as stated in the lab manual. The resulting discrepancy is a primary source of scale effects.
- **Mach Number (Ma):** This parameter will be conserved for a given test point.
- **Advance Ratio (J):** This parameter will be conserved to set the propeller operating state.

1.2.2. Performance Indicators

The measured outputs of the experiment are presented as dimensionless performance indicators. The overall aerodynamic performance of the airframe is quantified by the lift, drag, and pitching moment coefficients (C_L, C_D, and C_M). These coefficients normalize the measured aerodynamic forces and moments using the freestream dynamic pressure and the model's reference geometry, specifically the wing area (S) and mean aerodynamic chord ($\bar{c}$). Similarly, the performance of the propulsion system is characterized by the propeller thrust coefficient (C_T). This normalization is based on the fluid density, rotational speed (n), and propeller diameter (D), allowing for a consistent characterization of thrust performance across all flight speeds.

2. Measurement Techniques

This chapter details the different experimental techniques selected to address the measurement challenge. The experiment's results will be measured with the use of an external balance for aerodynamic load measurements and an inflow microphone for acoustic characterization. Furthermore, if there was enough time, a stethoscope would be used as an additional measurement technique.

2.1. Aerodynamic Measurements (External Balance)

Aerodynamic forces and moments acting on the aircraft will be measured using a 6-component external balance. The balance measures forces (F_x, F_y, F_z) and moments (M_x, M_y, M_z) along the balance reference frame axes, as illustrated in Figure 2.1.

Wind-off tare measurements must be taken to account for the model's weight and initial offsets for all model attitudes. These wind-off measurements need to be subtracted from the data during post-processing. Additionally, since the propellers do not have measuring devices, the forces acting on the propeller need to be estimated during post-processing. From this measurement technique the non-dimensional aerodynamic coefficients (C_L, C_D, C_m, etc.) are obtained.

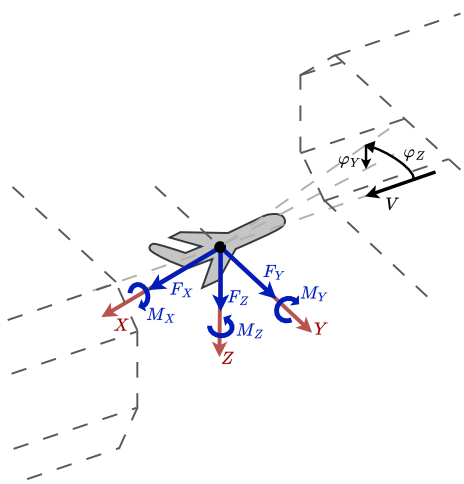

Figure 2.1: Balance axis system definition.

2.2. Acoustic Measurements (In-flow Microphone)

Before the lab exercise begins, the supervisor will calibrate an in-flow microphone. This measuring device returns an analog voltage signal as output, which through the supervisor's calibration is converted into an output pressure.

Finally, an optional stethoscope is provided for the lab exercise if there was enough time. The purpose of this flexibly positioned microphone is to auralize the flowfield. This serves as a qualitative way of distinguishing between laminar, turbulent and separated flow sections.

3. Approach to estimate propeller thrust from external balance data

3.1. Thrust Estimation and Data Reduction

3.1.1. Measurement Challenge

During wind tunnel runs, the thrust (T) cannot be measured directly because the propellers on the VGM scale model are not equipped with rotating shaft balances. Consequently, the thrust must be estimated. The external balance measures the net longitudinal force acting on the entire model, which is a superposition of the aerodynamic drag and the propulsive force.

To isolate the aerodynamic coefficients of the airframe (the primary objective), a specific data reduction procedure is applied.

3.1.2. Principle of Superposition

The external six-component balance measures the total longitudinal force (F_x). The relationship between the force measured by the balance, the airframe drag, and the propeller thrust is defined as:

$$F_{x,\text{balance}} = -D_{\text{airframe}} + T_{\text{total}} \tag{3.1}$$

Where:

- $F_{x,\text{balance}}$ is the raw force measured by the balance (after wind-off tare correction).
- D_{airframe} is the aerodynamic drag of the fuselage, wing, and empennage (including interference effects).
- T_{total} is the combined thrust generated by both propellers ($T_{\text{total}} = N_{\text{props}} \cdot T_{\text{single}}$).

For **Challenge 4** (braking/windmilling regime), the thrust T becomes negative (acting in the same direction as drag). The equation remains valid, but T_{total} will be a negative value, effectively increasing the magnitude of the rearward force measured by the balance.

3.1.3. Thrust Calculation Procedure

To solve for D_{airframe}, the thrust T_{total} is estimated using Blade Element Momentum (BEM) simulation fits. The procedure for each data point is as follows:

Using the measured free-stream velocity (V_∞) and the commanded propeller rotational speed (n in revolutions per second), the advance ratio is calculated:

$$J = \frac{V_\infty}{n \cdot D} \tag{3.2}$$

where $D = 0.2032\,\text{m}$ is the propeller diameter.

The thrust coefficient is determined using the polynomial equation derived from BEM simulations (specifically for pitch angle $\beta_{0.75} = 45°$).

For this experiment, the relationship is modelled as:

$$C_T(J) = -0.0051J^4 + 0.0959J^3 - 0.5888J^2 + 1.0065J - 0.1353 \tag{3.3}$$

Solving the equation above provides four roots: $J_1 \approx 2.5004$, $J_2 \approx 0.1467$ and $J_{3,4} \approx 8.078 \pm 2.656i$. Since $J_{3,4}$ are imaginary roots these cannot be achieved. Furthermore, J_2 is smaller than 1.6 which is the minimum J value possible. Thus, the only possible value for J at zero thrust is $J \approx 2.37$. Hence, using Equation 3.2 and rearranging for n.

$$n_{C_T=0} = \frac{V_\infty}{J \cdot D} = \frac{40}{2.5004 \cdot 0.2032} = 78.73Hz$$

For the braking challenge (4), the propeller should operate in the high-J range where the calculated C_T is negative.

Finally, the physical thrust force is calculated using the ambient density (ρ) measured in the test section and $N_{props} = 2$:

$$T_{\text{calc}} = C_T(J) \cdot \rho \cdot n^2 \cdot D^4 \cdot N_{\text{props}} \tag{3.4}$$

4. Description of Limitations of the Test Setup

Experimental simulations in the Low-Turbulence Tunnel (LTT) are subject to physical scale effects and operational constraints that affect data interpretation. These limitations are categorized as follows:

- **Operational and Performance Limits:**
  - **Velocity:** Maximum freestream velocity is limited to $40\,\text{m}\,\text{s}^{-1}$ due to structural limits and motor power output.

  - **Attitude:** Safe operation is limited to $\alpha \in [-5, +14]^\circ$ and $\beta \in [-10, +10]^\circ$.
  - **Propulsion:** The lowest achievable advance ratio is $J = 1.6$. Additionally, fixed blade pitch (45°) prevents simultaneous matching of thrust and power coefficients (C_T, C_P).

- **Similarity and Scaling Challenges:**
  - **Reynolds Number (Re):** Tunnel Re (2.7×10^6) is an order of magnitude lower than flight (27×10^6). Artificial transition strips are used to mitigate resulting boundary layer and stall discrepancies.
  - **Power Density:** Due to cubic scaling laws, model engine simulators require significantly higher power density than full-scale equivalents.

- **Facility and Setup Constraints:**
  - **Wall Interference:** Closed walls force streamlines, requiring boundary corrections for lift, solid blockage, and wake blockage.
  - **Acoustics:** Untreated hard walls cause reflections that contaminate noise measurements.
  - **Support Interference:** Struts modify local flow and introduce vortex shedding noise, which must be removed via tare measurements.

- **Instrumentation and Logistics:**
  - **Thrust:** Lacking individual load cells, thrust is derived by comparing power-on data to propeller-off reference sets.
  - **Resolution:** A single inflow microphone limits spatial resolution and prevents source localization.
  - **Downtime:** Manual changes to control surfaces or propellers incur ≈ 10 minutes of downtime per change.

5. Description of the Test and Test matrix with time planing

5.1. Objective variables

There are certain variables that need to be explored in this experiment in order to quantify the effects of propeller operation at negative thrust to brake the aircraft during approach. The main parameters of interest are; longitudinal stability, elevator effectiveness and control

power and aerodynamic performance in trimmed conditions. Furthermore, the experiment should quantify; the effects of boundary corrections on performance parameters, the effects of Reynolds number on performance parameters and the effects of operating conditions on the propeller noise emissions.

5.2. Selected data outputs products enabled by this matrix

Longitudinal stability

Longitudinal stability is obtained from the slope of the pitching-moment curve, which states that the moment coefficient slope with respect to the angle of attack should be negative when the moment coefficient is 0:

$$C_m = 0 \quad \rightarrow \quad \frac{dC_m}{d\alpha} < 0. \tag{5.1}$$

In the experiment, longitudinal stability is obtained by performing angle-of-attack sweeps at fixed freestream velocity, motor state, and elevator deflection. The pitching-moment coefficient is measured as a function of α, and the slope $\mathrm{d}C_m/\mathrm{d}\alpha$ is evaluated.

Elevator effectiveness and control power

Elevator effectiveness is defined as the change in moment coefficient based on the change in elevator deflection.

$$C_{m_{\delta_e}} = \frac{dC_m}{d\delta_e} \tag{5.2}$$

Control power can be estimated as the following, where $\delta_{e,\max}$ is the maximum deflection of the elevator:

$$\Delta C_m = C_{m_{\delta_e}} \delta_{e,\max} = \Delta C_{m_{max}} \tag{5.3}$$

In the experiment, elevator effectiveness is obtained from paired measurements at $\delta_e = -10°$ and $\delta_e = +10°$ under identical conditions. The maximum measured difference in pitching-moment coefficient between these two deflections is used to determine the available elevator control power.

Aerodynamic performance in trimmed conditions

The following variables will be explored in trimmed conditions, where the trim conditions are found at the point where the moment coefficient is 0.

$$\delta_{e,trim} \rightarrow C_m = 0 \quad C_{L,trim}, C_{D,trim}, C_{M,trim} \tag{5.4}$$

In practice, the trim elevator deflection is obtained by interpolating between the two measured elevator deflections for which the pitching-moment coefficient changes sign. The corresponding lift, drag, and moment coefficients are then evaluated at this trim condition.

Boundary corrections

Boundary corrections are applied in post-processing. The corrected aerodynamic coefficients are obtained by applying the prescribed tunnel boundary correction methodology explained in Chapter 4

Reynolds-number effect

The Reynolds-number effect is quantified by the inclusion of two free-stream velocities at identical of angle of attack, motor state, and control surface deflections.

Acoustic evaluation

Acoustic measurements are taken at a representative approach attitude ($\alpha = 8°$) for both motor states at 40 m/s:

$$V_\infty = 40\,\text{m/s}, \quad \alpha = 8°, \quad \beta = 0°, \quad \text{Motor} \in \{\text{OFF}, \text{BRAKING}(J{=}3.5)\}.$$

In the experiment, these measurements are obtained at dedicated operating points with extended sampling time to ensure a good noise spectra.

5.3. Test matrix

A full factorial test matrix is carried out in the wind tunnel. To reduce the number of configuration changes, the measurements are grouped by elevator deflection, with Block A at −10°, Block B at +10° and Block C at 0°. These deflections represent of the large elevator angles expected during landing. Although measurements at zero elevator deflection could give a good estimate of trim conditions, the chosen elevator blocks already allow elevator effectiveness, control power, and trimmed aerodynamic performance to be determined through interpolation, so an elevator deflection of +0° was not chosen. Nevertheless, a some experiments are run at the end to evaluate if interpolating the results is a good approximation. A small number of measurement points are repeated to check repeatability.

The selected sampling time per data point is selected to be 6[s] for a normal measurement and 15[s] for an acoustic measurement. Furthermore, to change angle of attack 15 [s] are used, and to change velocity 75 [s] are used. A 15 [min] tunnel start is selected and 10 [min] are allocated to change the elevator angle. In the Table A.1 the full test matrix can be seen.

5.4. Contingency plan

The selected test matrix has an approximately 39 [min] extra available in case that some measurements require extra time. Furthermore, if there were no need for that extra time, the optionally provided stethoscope would be used to provide insights on the flow's behaviour around certain areas of interest, such as near elevator etc.

A. Appendix

Table A.1: Wind-tunnel test matrix where: V_∞ is the free-stream velocity [m/s], M1 and M2 are the propeller speeds [Hz], α and β are the angle of attack and sideslip [deg], δ_r and δ_e are the rudder and elevator deflections [deg], t_s is the sampling time [s], t_e the time per data point [s], and t_a the accumulated time [min].

DPN	Block	V_∞	M1	M2	α	β	δ_r	δ_e	t_s	t_e	t_a	Notes
–	–	–	–	–	–	–	–	–	–	900	15.00	familiarisation + tunnel start
1	A (−10°)	0	OFF	OFF	-2	0	0	-10	6	36	15.60	wind-off calibrate
2	A (−10°)	0	OFF	OFF	0	0	0	-10	6	21	15.95	wind-off calibrate
3	A (−10°)	0	OFF	OFF	4	0	0	-10	6	21	16.30	wind-off calibrate
4	A (−10°)	0	OFF	OFF	8	0	0	-10	6	21	16.65	wind-off calibrate
5	A (−10°)	0	OFF	OFF	12	0	0	-10	6	21	17.00	wind-off calibrate
6	A (−10°)	40	OFF	OFF	12	0	0	-10	6	96	18.60	velocity change
7	A (−10°)	40	OFF	OFF	8	0	0	-10	6	21	18.95	aero (OFF)
8	A (−10°)	40	OFF	OFF	4	0	0	-10	6	21	19.30	aero (OFF)
9	A (−10°)	40	OFF	OFF	0	0	0	-10	15	45	20.05	aero + acoustic
10	A (−10°)	40	OFF	OFF	-2	0	0	-10	6	21	20.40	aero (OFF)
11	A (−10°)	40	78.7	78.7	-2	0	0	-10	6	36	21.00	motor change (BRK)
12	A (−10°)	40	78.7	78.7	0	0	0	-10	6	21	21.35	aero (BRK)
13	A (−10°)	40	78.7	78.7	4	0	0	-10	6	21	21.70	aero (BRK)
14	A (−10°)	40	78.7	78.7	8	0	0	-10	15	45	22.45	aero + acoustic
15	A (−10°)	40	78.7	78.7	12	0	0	-10	6	21	22.80	aero (BRK)
16	A (−10°)	20	OFF	OFF	12	0	0	-10	6	96	24.40	velocity change
17	A (−10°)	20	OFF	OFF	8	0	0	-10	6	21	24.75	aero (OFF)
18	A (−10°)	20	OFF	OFF	4	0	0	-10	6	21	25.10	aero (OFF)
19	A (−10°)	20	OFF	OFF	0	0	0	-10	6	21	25.45	aero (OFF)
20	A (−10°)	20	OFF	OFF	-2	0	0	-10	6	21	25.80	aero (OFF)
21	A (−10°)	20	39.4	39.4	-2	0	0	-10	6	36	26.40	motor change (BRK)
22	A (−10°)	20	39.4	39.4	0	0	0	-10	6	21	26.75	aero (BRK)
23	A (−10°)	20	39.4	39.4	4	0	0	-10	6	21	27.10	aero (BRK)
24	A (−10°)	20	39.4	39.4	8	0	0	-10	6	21	27.45	aero (BRK)
25	A (−10°)	20	39.4	39.4	12	0	0	-10	6	21	27.80	aero (BRK)

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DPN	Block	V_∞	M1	M2	α	β	δ_r	δ_e	t_s	t_e	t_a	Notes
26	A (−10°)	40	78.7	78.7	8	0	0	-10	6	36	28.40	replicate
–	–	–	–	–	–	–	–	–	–	600	38.40	elevator change
27	B (+10°)	0	OFF	OFF	-2	0	0	10	6	36	39.00	wind-off calibrate
28	B (+10°)	0	OFF	OFF	0	0	0	10	6	21	39.35	wind-off calibrate
29	B (+10°)	0	OFF	OFF	4	0	0	10	6	21	39.70	wind-off calibrate
30	B (+10°)	0	OFF	OFF	8	0	0	10	6	21	40.05	wind-off calibrate
31	B (+10°)	0	OFF	OFF	12	0	0	10	6	21	40.40	wind-off calibrate
32	B (+10°)	40	OFF	OFF	12	0	0	10	6	96	42.00	velocity change
33	B (+10°)	40	OFF	OFF	8	0	0	10	6	21	42.35	aero (OFF)
34	B (+10°)	40	OFF	OFF	4	0	0	10	6	21	42.70	aero (OFF)
35	B (+10°)	40	OFF	OFF	0	0	0	10	15	45	43.45	aero + acoustic
36	B (+10°)	40	OFF	OFF	-2	0	0	10	6	21	43.80	aero (OFF)
37	B (+10°)	40	78.7	78.7	-2	0	0	10	6	36	44.40	motor change (BRK)
38	B (+10°)	40	78.7	78.7	0	0	0	10	6	21	44.75	aero (BRK)
39	B (+10°)	40	78.7	78.7	4	0	0	10	6	21	45.10	aero (BRK)
40	B (+10°)	40	78.7	78.7	8	0	0	10	15	45	45.85	aero + acoustic
41	B (+10°)	40	78.7	78.7	12	0	0	10	6	21	46.20	aero (BRK)
42	B (+10°)	20	OFF	OFF	12	0	0	10	6	96	47.80	velocity change
43	B (+10°)	20	OFF	OFF	8	0	0	10	6	21	48.15	aero (OFF)
44	B (+10°)	20	OFF	OFF	4	0	0	10	6	21	48.50	aero (OFF)
45	B (+10°)	20	OFF	OFF	0	0	0	10	6	21	48.85	aero (OFF)
46	B (+10°)	20	OFF	OFF	-2	0	0	10	6	21	49.20	aero (OFF)
47	B (+10°)	20	39.4	39.4	-2	0	0	10	6	36	49.80	motor change (BRK)
48	B (+10°)	20	39.4	39.4	0	0	0	10	6	21	50.15	aero (BRK)
49	B (+10°)	20	39.4	39.4	4	0	0	10	6	21	50.50	aero (BRK)
50	B (+10°)	20	39.4	39.4	8	0	0	10	6	21	50.85	aero (BRK)
51	B (+10°)	20	39.4	39.4	12	0	0	10	6	21	51.20	aero (BRK)
52	B (+10°)	40	78.7	78.7	8	0	0	10	6	36	51.80	replicate
–	–	–	–	–	–	–	–	–	–	600	61.80	elevator change
53	C (0°)	0	OFF	OFF	-2	0	0	-10	6	21	62.15	no elevator calibrate
54	C (0°)	0	OFF	OFF	0	0	0	-10	6	21	62.50	no elevator calibrate
55	C (0°)	0	OFF	OFF	4	0	0	-10	6	21	62.85	no elevator calibrate
56	C (0°)	0	OFF	OFF	8	0	0	-10	6	21	63.20	no elevator calibrate
57	C (0°)	0	OFF	OFF	12	0	0	-10	6	21	63.55	no elevator calibrate
58	C (0°)	40	OFF	OFF	-2	0	0	-10	6	21	63.90	no elevator calibrate
59	C (0°)	40	OFF	OFF	0	0	0	-10	6	21	64.25	no elevator calibrate

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DPN	Block	V_∞	M1	M2	α	β	δ_r	δ_e	t_s	t_e	t_a	Notes
60	C (0°)	40	OFF	OFF	4	0	0	-10	6	21	64.60	no elevator calibrate
61	C (0°)	40	OFF	OFF	8	0	0	-10	6	21	64.95	no elevator calibrate
62	C (0°)	40	OFF	OFF	12	0	0	-10	6	21	65.30	no elevator calibrate

C Statement of Fair Conduct

We hereby declare that:

- I have fairly contributed, in the measure of at least ≈20%, to the generation of all the parts of the delivered material,
- All the delivered material is the result of the work performed by my team and no others,
- I am aware of the consequences of committing fraud[5] (i.e. piggybacking, copying, fiddling with data and plagiarism).

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[5] https://www.tudelft.nl/en/student/my-study-me/rules-guidelines-and-participation/fraud-plagiarism